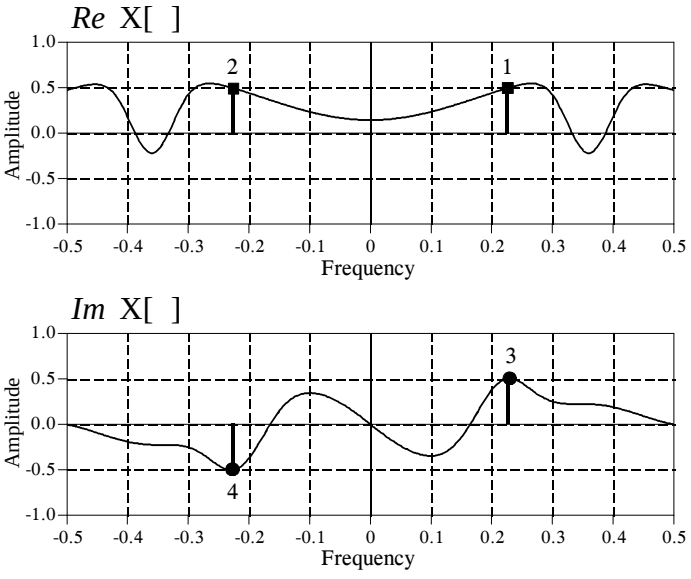


FIGURE 31-1
Complex frequency spectrum. These curves correspond to an entirely real time domain signal, because the real part of the spectrum has an *even* symmetry, and the imaginary part has an *odd* symmetry. The two square markers in the real part correspond to a cosine wave with an amplitude of one, and a frequency of 0.23. The two round markers in the imaginary part correspond to a sine wave with an amplitude of one, and a frequency of 0.23.

Figure 31-1 illustrates the complex DFT's frequency spectrum. This figure assumes the time domain is entirely *real*, that is, its imaginary part is zero. We will discuss the idea of imaginary time domain signals shortly. There are two common ways of displaying a complex frequency spectrum. As shown here, zero frequency can be placed in the center, with positive frequencies to the right and negative frequencies to the left. This is the best way to *think* about the complete spectrum, and is the *only* way that an aperiodic spectrum can be displayed.

The problem is that the spectrum of a discrete signal is *periodic* (such as with the DFT and the DTFT). This means that everything between -0.5 and 0.5 repeats itself an infinite number of times to the left and to the right. In this case, the spectrum between 0 and 1.0 contains the same information as from -0.5 to 0.5. When graphs are made, such as Fig. 31-1, the -0.5 to 0.5 convention is usually used. However, many equations and programs use the 0 to 1.0 form. For instance, in Eqs. 31-5 and 31-6 the frequency index, k, runs from 0 to N-1 (coinciding with 0 to 1.0). However, we could write it to run from $-N/2$ to $N/2$-1 (coinciding with -0.5 to 0.5), if we desired.

Using the spectrum in Fig. 31-1 as a guide, we can examine how the inverse complex DFT reconstructs the time domain signal. The inverse complex DFT, written in polar form, is given by:

EQUATION 31-7
The inverse complex DFT. This is matching equation to the forward complex DFT in Eq. 31-5.

$$x[n] = \sum_{k=0}^{N-1} X[k] e^{j 2\pi kn/N}$$